Approximating price series, part 3: MLP of t (tanh) and Fourier-feature MLP of t

Approximation study, 2020-01-02 to 2023-07-31 daily closes, three tickers. All numbers recomputed from the saved fits (approx_study/fits/*.json).

1. What is being measured

This is an in-sample APPROXIMATION task, not forecasting. Each model receives the entire close-price series of a stock, p(t), with t rescaled to [0, 1] (905 points for CBA, 901 for TSLA and LLY; prices in the file's whole-series Min-Max scale, 0 = lowest close of the period, 1 = highest), and must reproduce it as a function of t with a fixed number of parameters. The score is the RMSE and the maximum absolute error between the fitted curve and the actual series, and how both fall as the parameter budget grows (about 15, 30, 60, 120, 240). There is no train/test split and no prediction; in-sample fit says nothing about forecasting. For scale, the dotted line in the error plots is the one-day-ahead naive forecasting error of the same series (0.0201 CBA, 0.0255 TSLA, 0.0148 LLY).

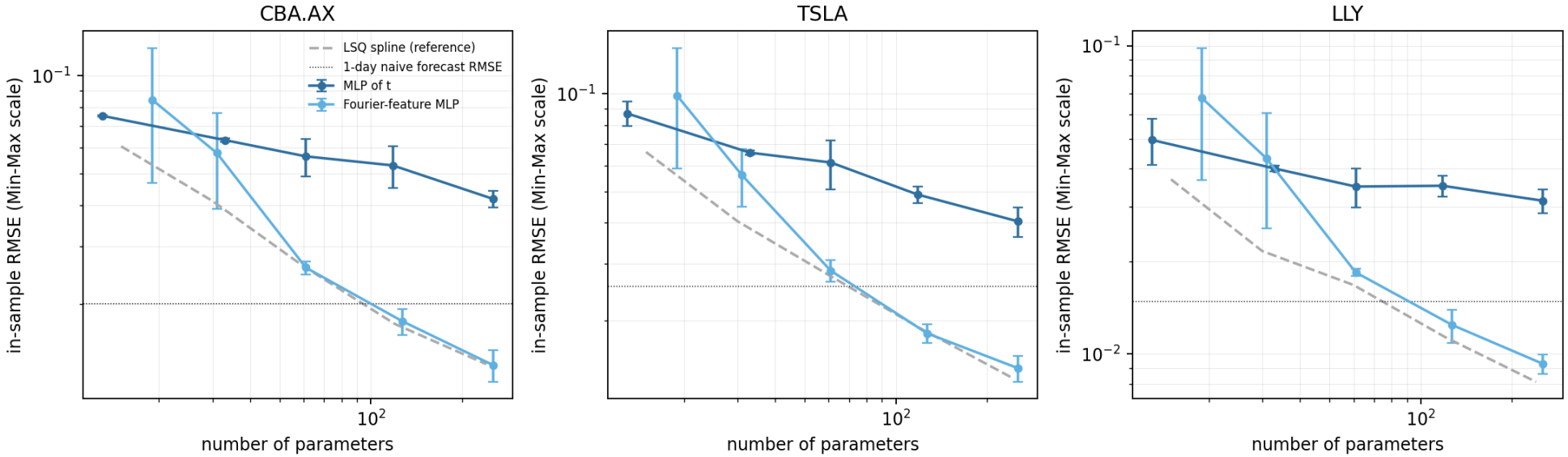
2. The two models in this document

MLP of t: 1 -> w -> w -> 1, tanh hidden units, linear output
Fourier-feature MLP: [sin(2 pi B t), cos(2 pi B t)] -> w -> w -> 1, same network
• Both are PyTorch MLPs with two hidden layers of width w and tanh units, trained full-batch with Adam (lr 0.01, cosine schedule) for 20,000 steps, keeping the best-loss weights. Three seeds (0, 1, 2).
• MLP of t: input is the scalar t. Parameters = w^2 + 4w + 1.
• Fourier-feature MLP: the identical network, but its input is a fixed random map of t into 2 n_freq sinusoids, frequencies B ~ N(0, 6^2) drawn once per seed and NOT trained (Tancik et al. 2020). Parameters = w^2 + (2 n_freq + 3) w + 1; the width w is reduced so the total matches the budget.
• The one thing that changes is the input basis. Tancik et al. prove a plain MLP of a coordinate learns low frequencies first and high frequencies extremely slowly (spectral bias); the Fourier map removes that.
• Training length matters for the plain MLP: 4,000 steps gave RMSE 0.074 on CBA at 61 parameters, 20,000 steps 0.048; all results below use 20,000 steps.

3. Settings per budget

target budget	15	30	60	120
MLP of t actual params	13	33	61	118
Fourier-feature MLP actual params	19	31	61	127
setting	MLP width w = 2; Fourier MLP (n_freq, w) = (2, 2)	MLP width w = 4; Fourier MLP (n_freq, w) = (2, 3)	MLP width w = 6; Fourier MLP (n_freq, w) = (4, 4)	MLP width w = 9; Fourier MLP (n_freq, w)

4. Error versus number of parameters (all tickers)



Points are the mean over seeds, bars the sample standard deviation across seeds (absent for deterministic models). Log-log axes: a straight line means error falls as a power of the parameter count.

Slope of log RMSE against log parameters, budgets 30-240 (how fast each model improves with size):

model	CBA.AX	TSLA	LLY	mean
MLP of t	-0.20	-0.25	-0.11	-0.18
Fourier-feature MLP	-0.69	-0.65	-0.71	-0.68

5. Every fit: full table

One row per fit (ticker, model, budget, seed). RMSE and max |error| in the ticker's Min-Max scale; seconds = wall-clock fit time.

ticker	model	budget	seed	params	RMSE	max |err|	seconds	setting
CBA.AX	mlp	15	0	13	0.0758	0.2554	13.0	w=2
CBA.AX	mlp	15	1	13	0.0748	0.2414	11.2	w=2
CBA.AX	mlp	15	2	13	0.0752	0.2421	11.2	w=2
CBA.AX	mlp	30	0	33	0.0637	0.2173	10.6	w=4
CBA.AX	mlp	30	1	33	0.0641	0.2106	10.3	w=4
CBA.AX	mlp	30	2	33	0.0624	0.1884	10.7	w=4
CBA.AX	mlp	60	0	61	0.0482	0.1924	11.1	w=6
CBA.AX	mlp	60	1	61	0.0626	0.2160	11.9	w=6
CBA.AX	mlp	60	2	61	0.0588	0.2011	12.0	w=6
CBA.AX	mlp	120	0	118	0.0493	0.1958	10.5	w=9
CBA.AX	mlp	120	1	118	0.0478	0.1977	11.1	w=9
CBA.AX	mlp	120	2	118	0.0621	0.2114	11.4	w=9
CBA.AX	mlp	240	0	253	0.0393	0.1297	12.3	w=14
CBA.AX	mlp	240	1	253	0.0440	0.1618	14.7	w=14
CBA.AX	mlp	240	2	253	0.0424	0.1403	12.9	w=14
CBA.AX	fmlp	15	0	19	0.0660	0.2293	10.9	n_freq=2, w=2, sigma=6
CBA.AX	fmlp	15	1	19	0.1272	0.6627	10.9	n_freq=2, w=2, sigma=6
CBA.AX	fmlp	15	2	19	0.0595	0.1903	10.2	n_freq=2, w=2, sigma=6
CBA.AX	fmlp	30	0	31	0.0544	0.2045	10.4	n_freq=2, w=3, sigma=6
CBA.AX	fmlp	30	1	31	0.0783	0.3452	10.4	n_freq=2, w=3, sigma=6
CBA.AX	fmlp	30	2	31	0.0411	0.1372	10.7	n_freq=2, w=3, sigma=6
CBA.AX	fmlp	60	0	61	0.0271	0.1054	14.0	n_freq=4, w=4, sigma=6
CBA.AX	fmlp	60	1	61	0.0249	0.0793	10.5	n_freq=4, w=4, sigma=6
CBA.AX	fmlp	60	2	61	0.0254	0.1052	11.5	n_freq=4, w=4, sigma=6
CBA.AX	fmlp	120	0	127	0.0195	0.0909	10.5	n_freq=6, w=6, sigma=6
CBA.AX	fmlp	120	1	127	0.0170	0.0800	10.5	n_freq=6, w=6, sigma=6

ticker	model	budget	seed	params	RMSE	max |err|	seconds	setting
CBA.AX	fmlp	120	2	127	0.0165	0.0781	11.1	n_freq=6, w=6, sigma=6
CBA.AX	fmlp	240	0	253	0.0128	0.0615	12.1	n_freq=8, w=9, sigma=6
CBA.AX	fmlp	240	1	253	0.0145	0.0751	14.1	n_freq=8, w=9, sigma=6
CBA.AX	fmlp	240	2	253	0.0116	0.0646	13.5	n_freq=8, w=9, sigma=6
TSLA	mlp	15	0	13	0.0930	0.2976	11.8	w=2
TSLA	mlp	15	1	13	0.0785	0.3023	11.6	w=2
TSLA	mlp	15	2	13	0.0890	0.3080	11.3	w=2
TSLA	mlp	30	0	33	0.0650	0.2473	11.8	w=4
TSLA	mlp	30	1	33	0.0654	0.2448	11.9	w=4
TSLA	mlp	30	2	33	0.0669	0.2549	11.4	w=4
TSLA	mlp	60	0	61	0.0530	0.2393	11.0	w=6
TSLA	mlp	60	1	61	0.0732	0.2521	10.3	w=6
TSLA	mlp	60	2	61	0.0578	0.2453	10.3	w=6
TSLA	mlp	120	0	118	0.0457	0.1785	10.5	w=9
TSLA	mlp	120	1	118	0.0510	0.2378	11.2	w=9
TSLA	mlp	120	2	118	0.0499	0.1982	11.2	w=9
TSLA	mlp	240	0	253	0.0356	0.1480	11.7	w=14
TSLA	mlp	240	1	253	0.0433	0.1695	11.4	w=14
TSLA	mlp	240	2	253	0.0422	0.1510	11.5	w=14
TSLA	fmlp	15	0	19	0.0700	0.2259	11.3	n_freq=2, w=2, sigma=6
TSLA	fmlp	15	1	19	0.1440	0.5914	11.4	n_freq=2, w=2, sigma=6
TSLA	fmlp	15	2	19	0.0816	0.3109	11.7	n_freq=2, w=2, sigma=6
TSLA	fmlp	30	0	31	0.0550	0.2344	11.8	n_freq=2, w=3, sigma=6
TSLA	fmlp	30	1	31	0.0681	0.3133	14.2	n_freq=2, w=3, sigma=6
TSLA	fmlp	30	2	31	0.0454	0.1924	10.4	n_freq=2, w=3, sigma=6
TSLA	fmlp	60	0	61	0.0289	0.1162	10.5	n_freq=4, w=4, sigma=6
TSLA	fmlp	60	1	61	0.0304	0.1423	10.2	n_freq=4, w=4, sigma=6
TSLA	fmlp	60	2	61	0.0261	0.1146	10.1	n_freq=4, w=4, sigma=6

ticker	model	budget	seed	params	RMSE	max |err|	seconds	setting
TSLA	fmlp	120	0	127	0.0195	0.0862	10.4	n_freq=6, w=6, sigma=6
TSLA	fmlp	120	1	127	0.0181	0.0745	10.7	n_freq=6, w=6, sigma=6
TSLA	fmlp	120	2	127	0.0171	0.0829	12.9	n_freq=6, w=6, sigma=6
TSLA	fmlp	240	0	253	0.0151	0.0807	11.4	n_freq=8, w=9, sigma=6
TSLA	fmlp	240	1	253	0.0150	0.0724	10.6	n_freq=8, w=9, sigma=6
TSLA	fmlp	240	2	253	0.0127	0.0497	11.0	n_freq=8, w=9, sigma=6
LLY	mlp	15	0	13	0.0586	0.2014	10.6	w=2
LLY	mlp	15	1	13	0.0421	0.1073	10.5	w=2
LLY	mlp	15	2	13	0.0478	0.1351	10.4	w=2
LLY	mlp	30	0	33	0.0408	0.1135	10.4	w=4
LLY	mlp	30	1	33	0.0391	0.1042	10.3	w=4
LLY	mlp	30	2	33	0.0400	0.1114	10.7	w=4
LLY	mlp	60	0	61	0.0379	0.1113	10.9	w=6
LLY	mlp	60	1	61	0.0379	0.1107	11.2	w=6
LLY	mlp	60	2	61	0.0291	0.0870	10.4	w=6
LLY	mlp	120	0	118	0.0343	0.1130	10.9	w=9
LLY	mlp	120	1	118	0.0382	0.1128	10.8	w=9
LLY	mlp	120	2	118	0.0330	0.0991	11.1	w=9
LLY	mlp	240	0	253	0.0302	0.1052	11.6	w=14
LLY	mlp	240	1	253	0.0347	0.1120	11.3	w=14
LLY	mlp	240	2	253	0.0294	0.0907	11.3	w=14
LLY	fmlp	15	0	19	0.0388	0.1119	10.5	n_freq=2, w=2, sigma=6
LLY	fmlp	15	1	19	0.1002	0.4593	10.5	n_freq=2, w=2, sigma=6
LLY	fmlp	15	2	19	0.0639	0.2470	10.2	n_freq=2, w=2, sigma=6
LLY	fmlp	30	0	31	0.0357	0.1160	12.0	n_freq=2, w=3, sigma=6
LLY	fmlp	30	1	31	0.0631	0.2697	10.1	n_freq=2, w=3, sigma=6
LLY	fmlp	30	2	31	0.0305	0.1091	10.6	n_freq=2, w=3, sigma=6
LLY	fmlp	60	0	61	0.0189	0.0694	10.6	n_freq=4, w=4, sigma=6

ticker	model	budget	seed	params	RMSE	max |err|	seconds	setting
LLY	fmlp	60	1	61	0.0183	0.0767	10.6	n_freq=4, w=4, sigma=6
LLY	fmlp	60	2	61	0.0180	0.0694	10.2	n_freq=4, w=4, sigma=6
LLY	fmlp	120	0	127	0.0139	0.0529	10.4	n_freq=6, w=6, sigma=6
LLY	fmlp	120	1	127	0.0124	0.0516	12.6	n_freq=6, w=6, sigma=6
LLY	fmlp	120	2	127	0.0109	0.0480	10.6	n_freq=6, w=6, sigma=6
LLY	fmlp	240	0	253	0.0097	0.0473	11.0	n_freq=8, w=9, sigma=6
LLY	fmlp	240	1	253	0.0097	0.0416	10.8	n_freq=8, w=9, sigma=6
LLY	fmlp	240	2	253	0.0085	0.0378	10.6	n_freq=8, w=9, sigma=6

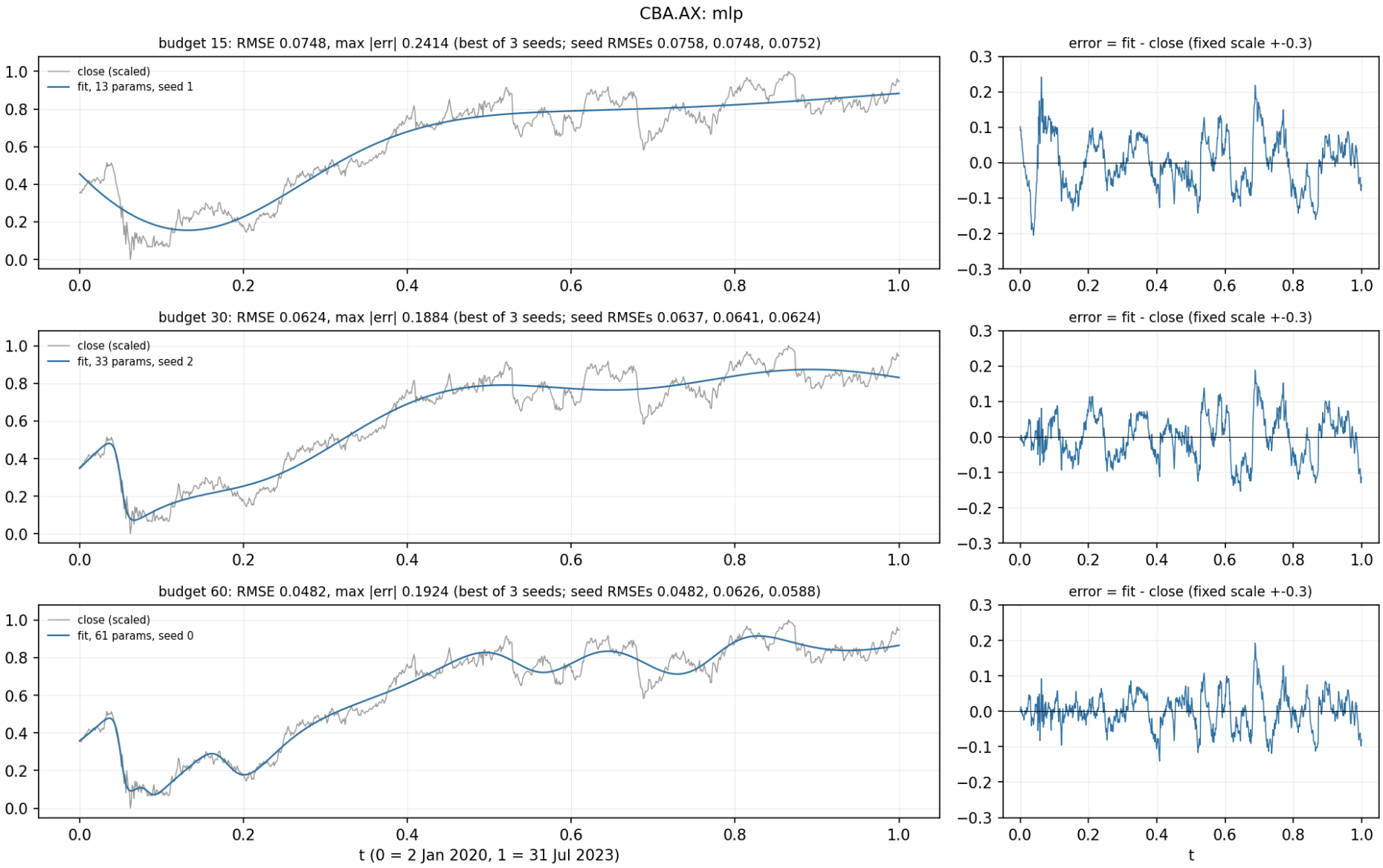
6. Summary per ticker and budget (mean +- sd over seeds)

ticker	budget	MLP of t RMSE	max |err|	params	Fourier-feature MLP RMSE	max |err|	params
CBA.AX	15	0.0753 +- 0.0005	0.2463	13	0.0842 +- 0.0373	0.3608	19
CBA.AX	30	0.0634 +- 0.0009	0.2054	33	0.0579 +- 0.0189	0.2290	31
CBA.AX	60	0.0566 +- 0.0074	0.2032	61	0.0258 +- 0.0012	0.0966	61
CBA.AX	120	0.0531 +- 0.0078	0.2016	118	0.0177 +- 0.0016	0.0830	127
CBA.AX	240	0.0419 +- 0.0024	0.1439	253	0.0130 +- 0.0014	0.0670	253
TSLA	15	0.0868 +- 0.0075	0.3026	13	0.0985 +- 0.0398	0.3760	19
TSLA	30	0.0658 +- 0.0010	0.2490	33	0.0562 +- 0.0114	0.2467	31
TSLA	60	0.0614 +- 0.0106	0.2456	61	0.0285 +- 0.0022	0.1244	61
TSLA	120	0.0489 +- 0.0028	0.2049	118	0.0182 +- 0.0012	0.0812	127
TSLA	240	0.0404 +- 0.0041	0.1562	253	0.0143 +- 0.0013	0.0676	253
LLY	15	0.0495 +- 0.0084	0.1479	13	0.0676 +- 0.0309	0.2728	19
LLY	30	0.0400 +- 0.0008	0.1097	33	0.0431 +- 0.0175	0.1649	31
LLY	60	0.0349 +- 0.0051	0.1030	61	0.0184 +- 0.0005	0.0718	61
LLY	120	0.0352 +- 0.0027	0.1083	118	0.0124 +- 0.0015	0.0508	127
LLY	240	0.0314 +- 0.0028	0.1026	253	0.0093 +- 0.0007	0.0422	253

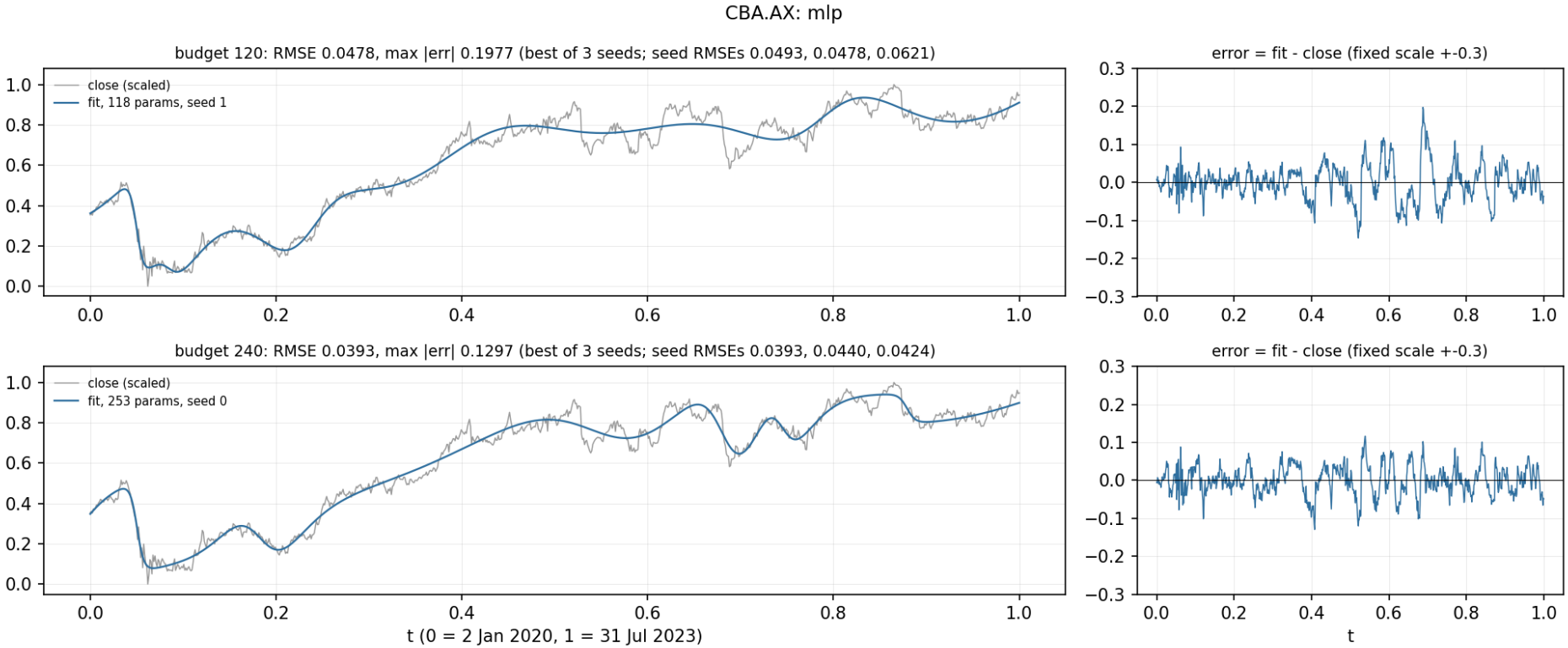
7. The fits themselves (following pages)

For every ticker, model and budget: left, the fitted curve over the actual series (best seed where there are seeds; all seed RMSEs in the title); right, the error fit - close on a fixed +-0.3 scale so panels are comparable. Vertical lines in spline panels are knot positions.

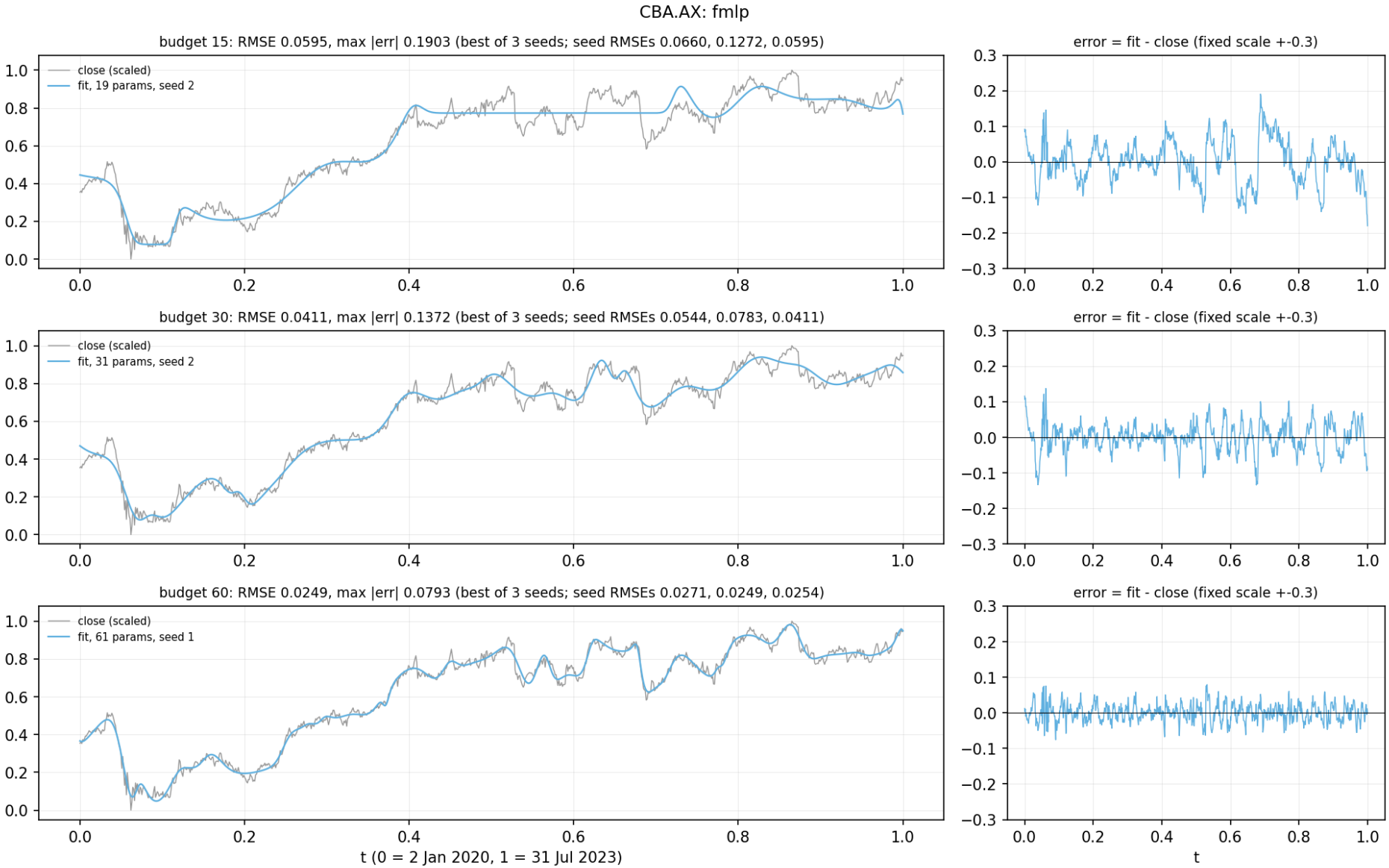
CBA.AX (Commonwealth Bank, benchmark series), MLP of t: 1 -> w -> w -> 1, tanh hidden units, linear output (budgets 15, 30, 60)



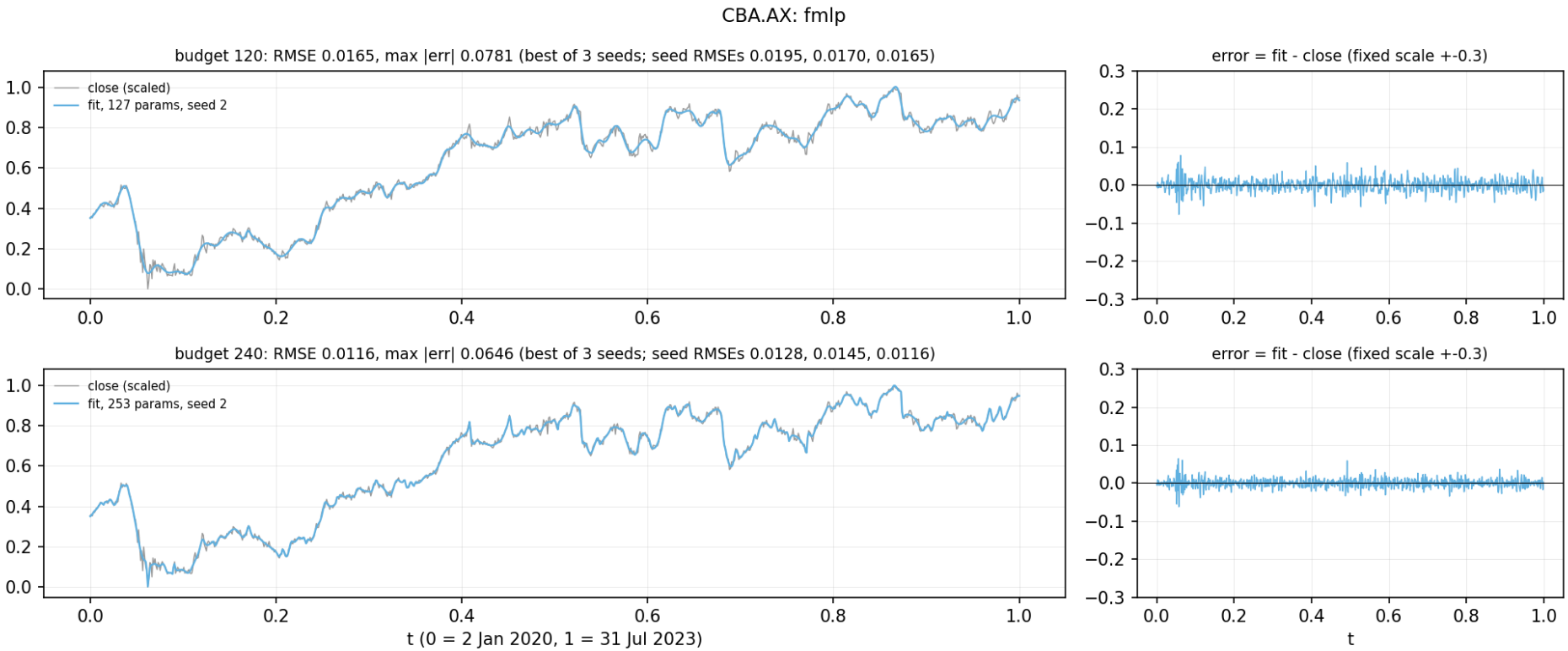
CBA.AX (Commonwealth Bank, benchmark series), MLP of t: 1 -> w -> w -> 1, tanh hidden units, linear output (budgets 120, 240)



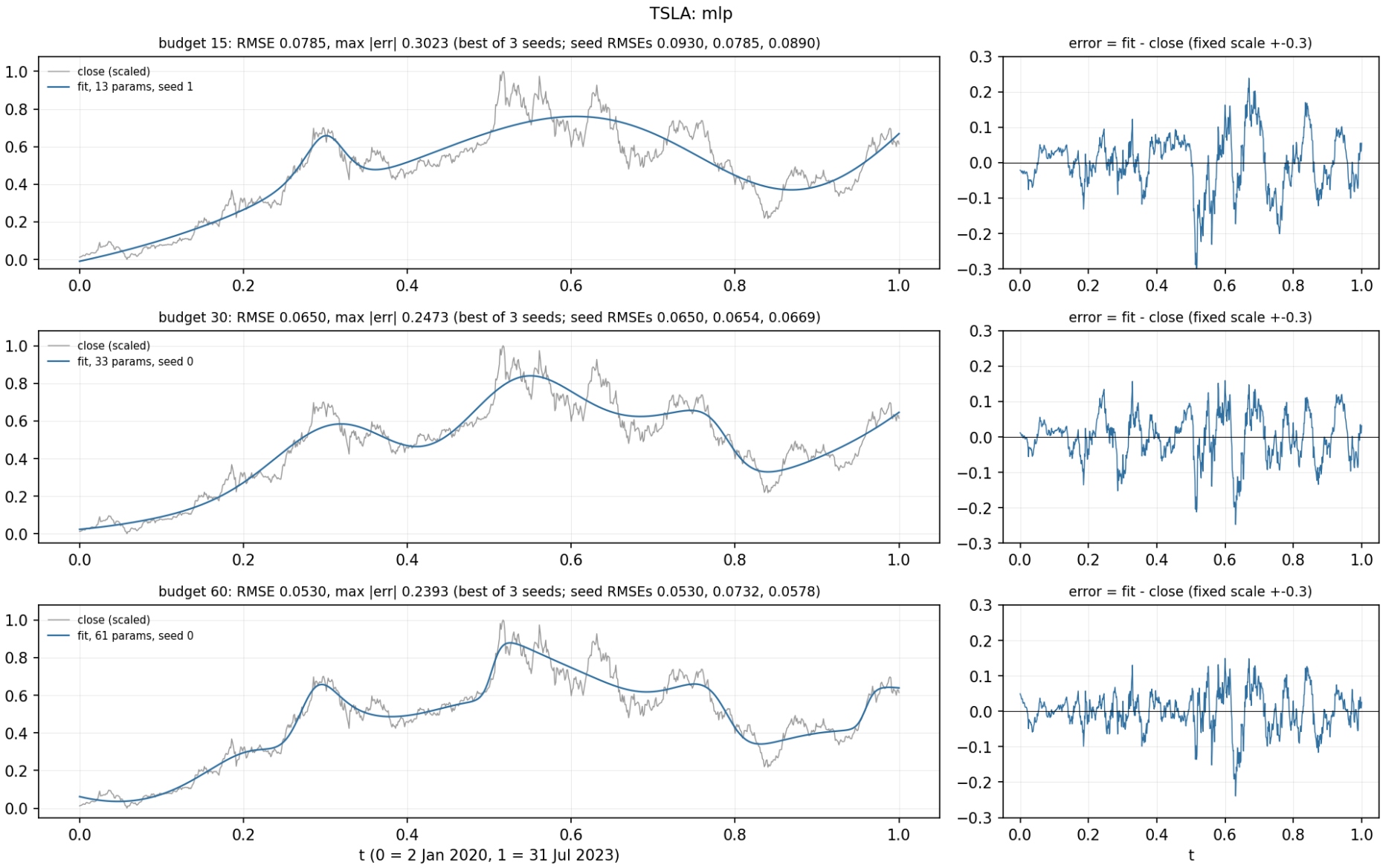
CBA.AX (Commonwealth Bank, benchmark series), Fourier-feature MLP: [sin(2 pi B t), cos(2 pi B t)] -> w -> w -> 1, same network (budgets 15, 30, 60)



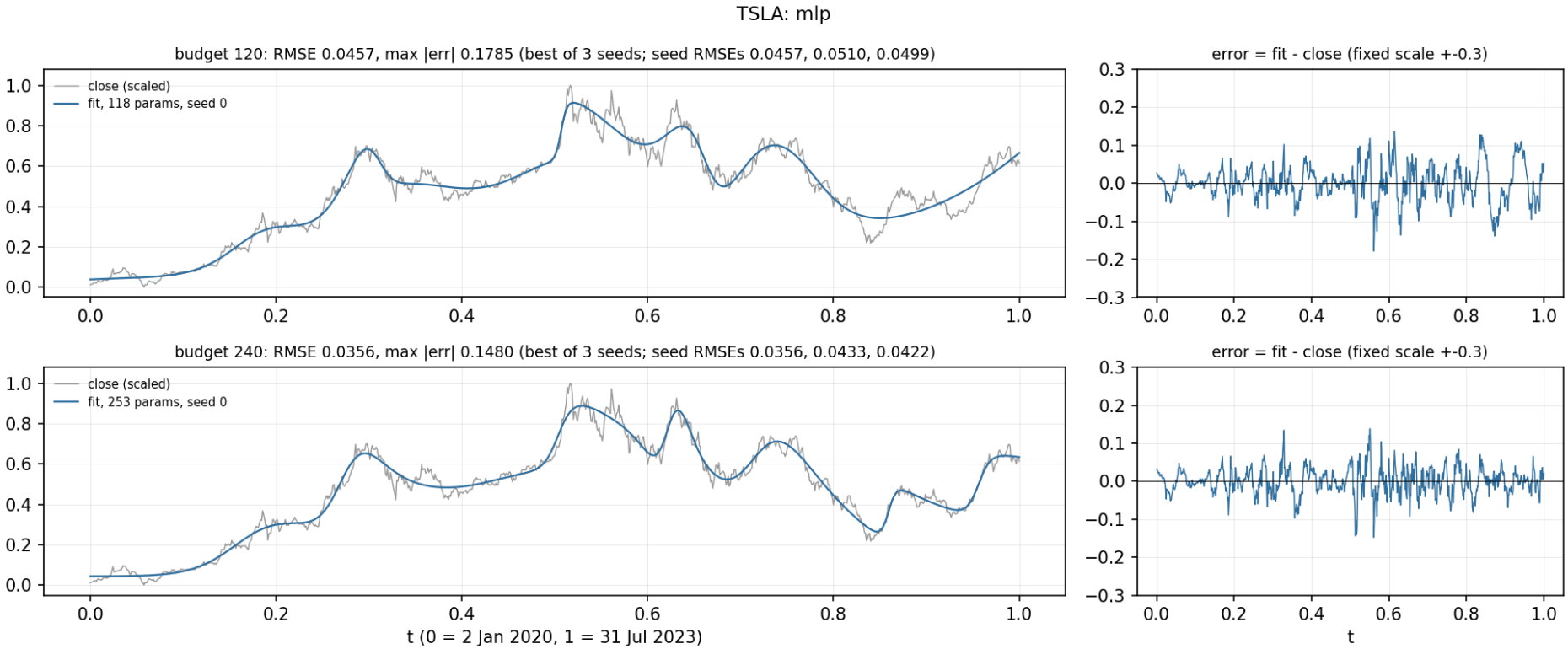
CBA.AX (Commonwealth Bank, benchmark series), Fourier-feature MLP: [sin(2 pi B t), cos(2 pi B t)] -> w -> w -> 1, same network (budgets 120, 240)



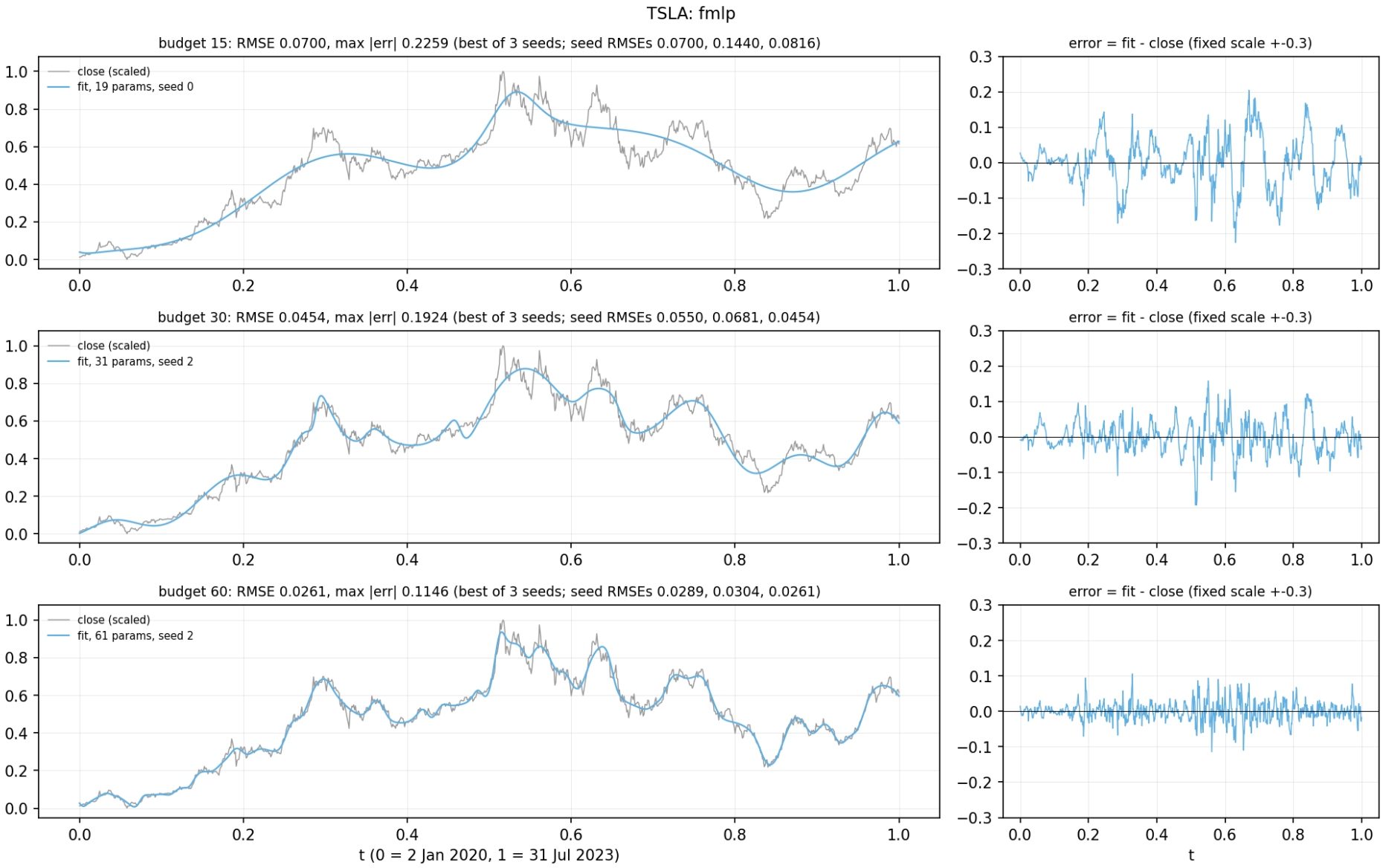
TSLA (Tesla, volatile), MLP of t: 1 -> w -> w -> 1, tanh hidden units, linear output (budgets 15, 30, 60)



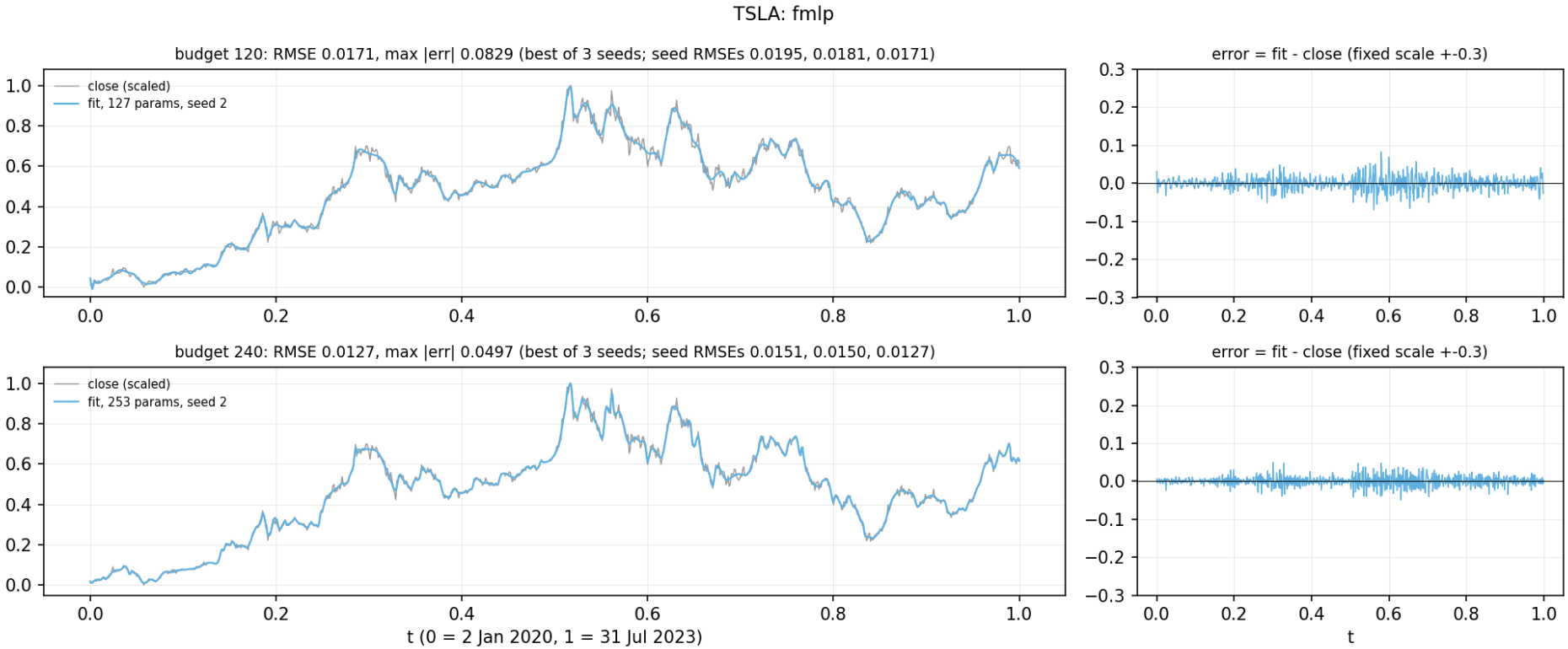
TSLA (Tesla, volatile), MLP of t: 1 -> w -> w -> 1, tanh hidden units, linear output (budgets 120, 240)



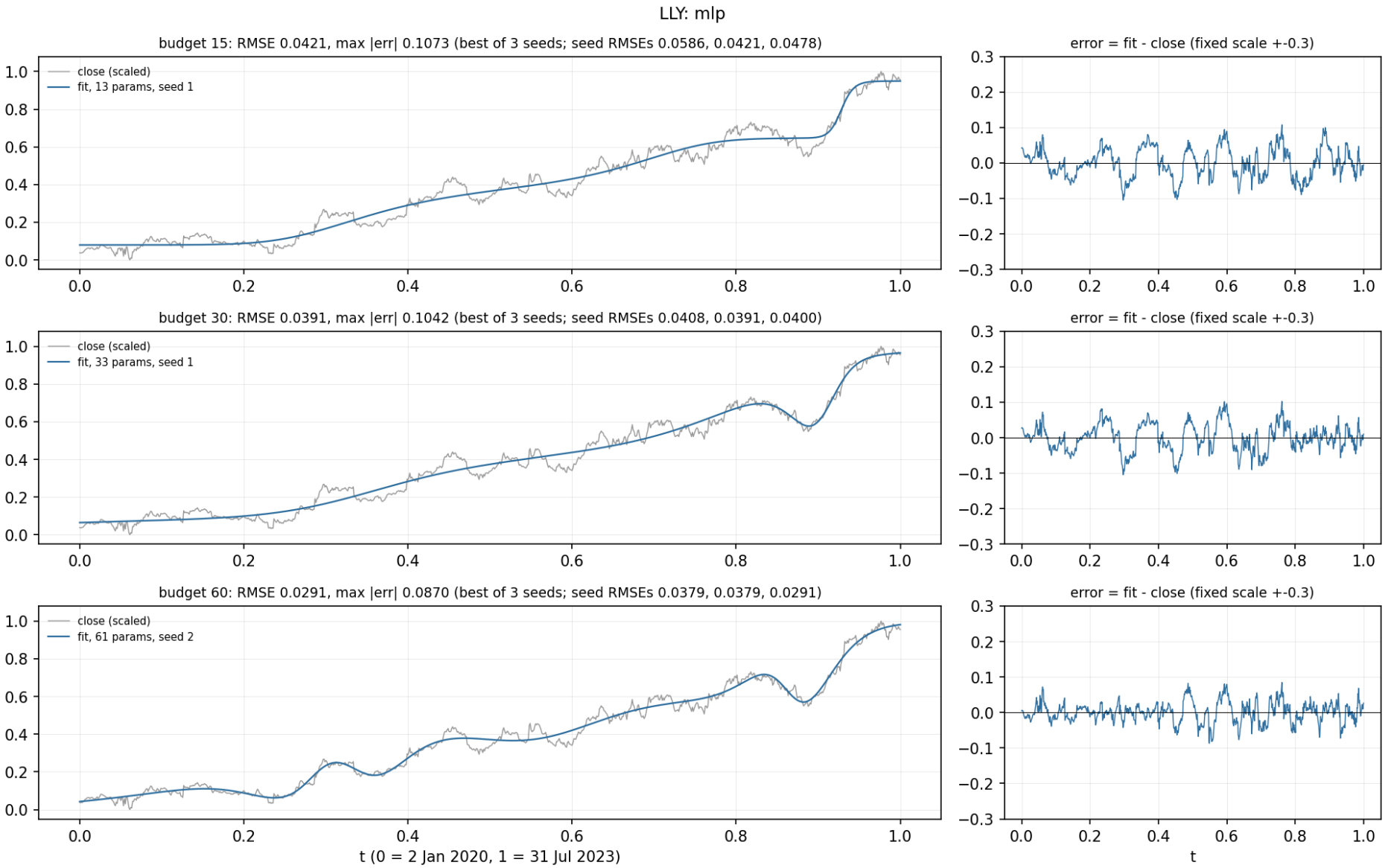
TSLA (Tesla, volatile), Fourier-feature MLP: [sin(2 pi B t), cos(2 pi B t)] -> w -> w -> 1, same network (budgets 15, 30, 60)



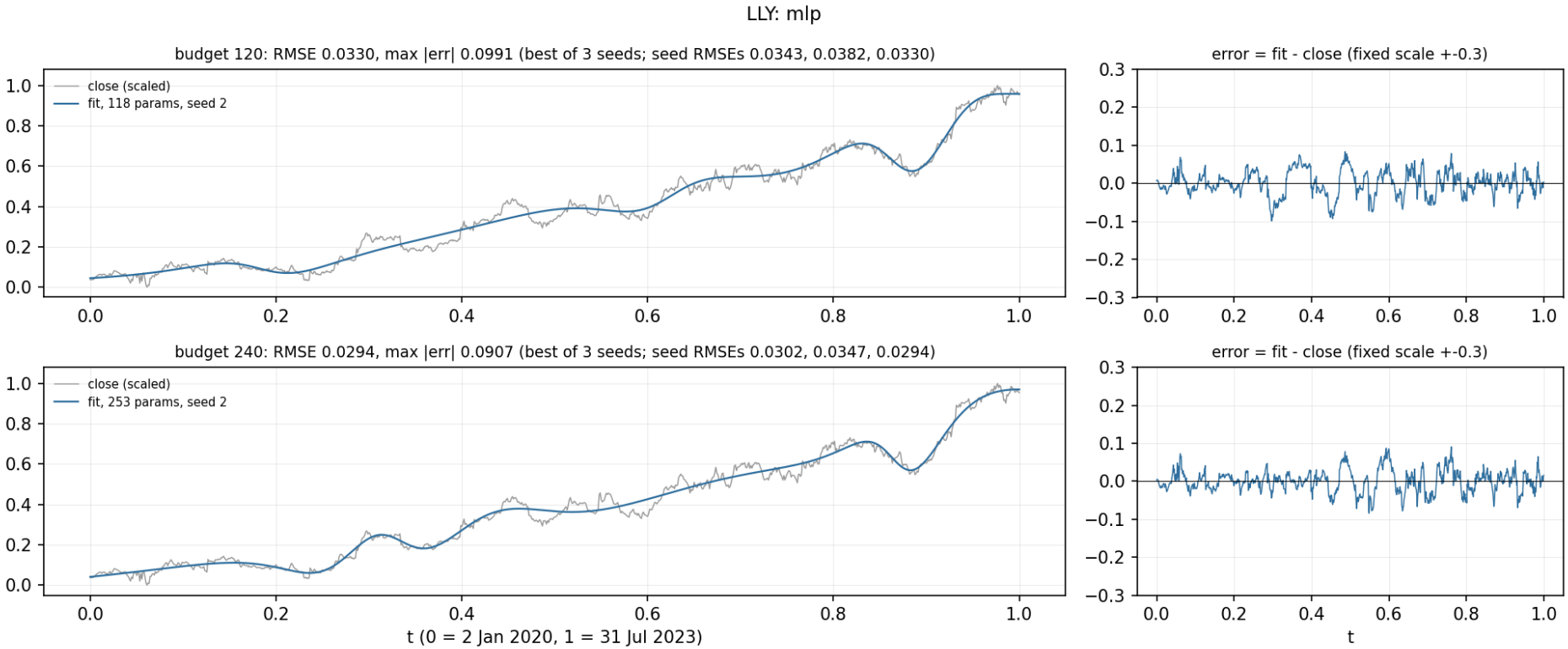
TSLA (Tesla, volatile), Fourier-feature MLP: [sin(2 pi B t), cos(2 pi B t)] -> w -> w -> 1, same network (budgets 120, 240)



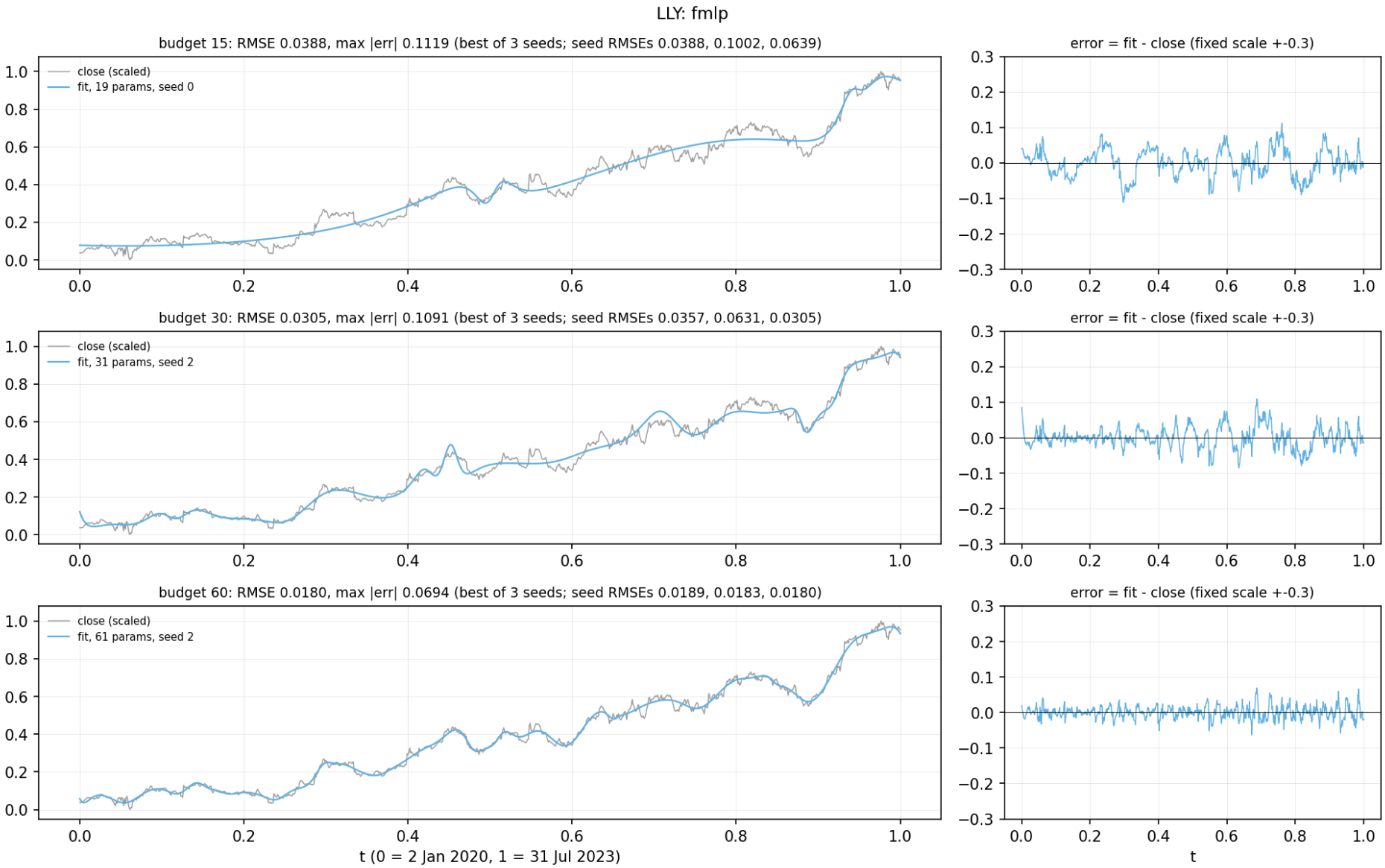
LLY (Eli Lilly, stable trend), MLP of t: 1 -> w -> w -> 1, tanh hidden units, linear output (budgets 15, 30, 60)



LLY (Eli Lilly, stable trend), MLP of t: 1 -> w -> w -> 1, tanh hidden units, linear output (budgets 120, 240)



LLY (Eli Lilly, stable trend), Fourier-feature MLP: [sin(2 pi B t), cos(2 pi B t)] -> w -> w -> 1, same network (budgets 15, 30, 60)



LLY (Eli Lilly, stable trend), Fourier-feature MLP: [sin(2 pi B t), cos(2 pi B t)] -> w -> w -> 1, same network (budgets 120, 240)

